Data Handling and Visualization

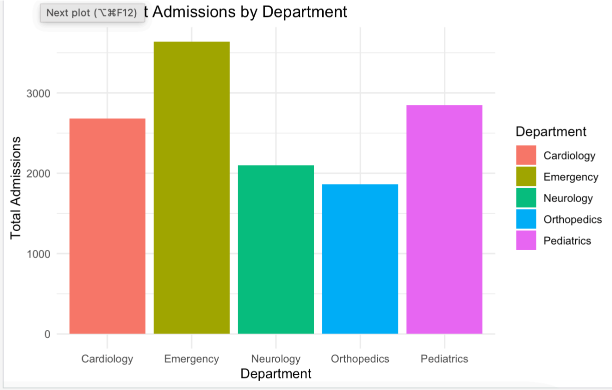
Class Activity - 3

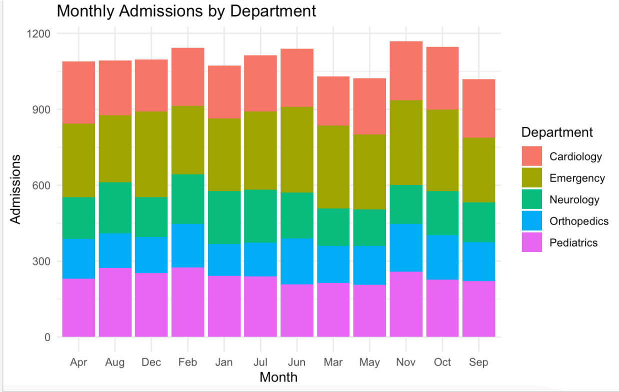
Date: 20/07/2026

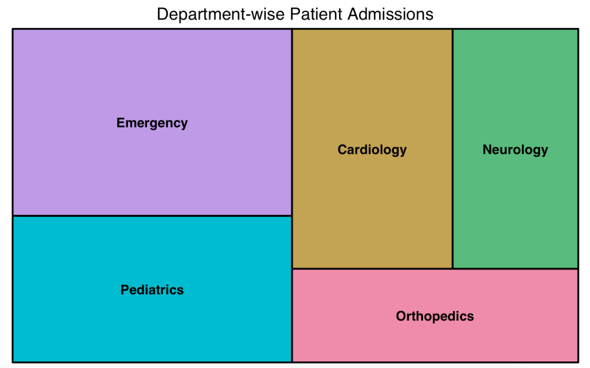
1. Hospital Patient Admission Analysis

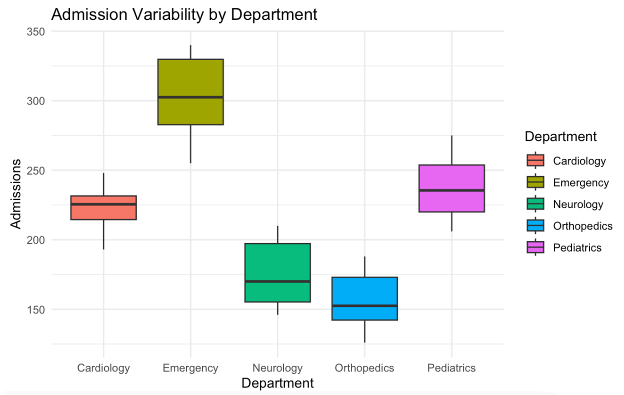
R program

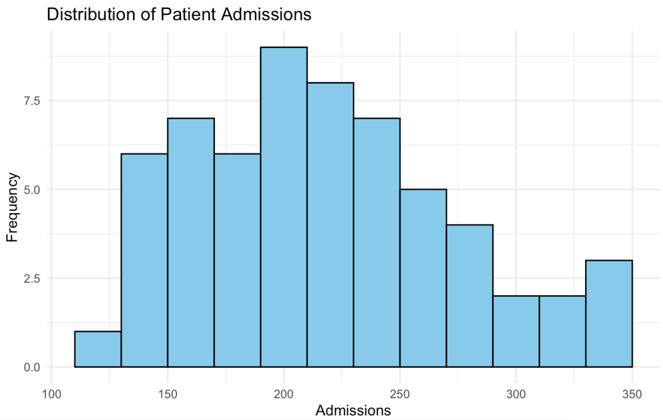
```
library(ggplot2)
library(dplyr)
library(treemap)

set.seed(123)

hospital <- data.frame(
  Department = rep(c("Cardiology", "Neurology", "Pediatrics", "Orthopedics", "Emergency"), each = 12),
  Month = rep(month.abb, 5),
  Admissions = c(
    sample(180:250, 12),
    sample(140:210, 12),
    sample(200:280, 12),
    sample(120:190, 12),
    sample(250:350, 12)
  )
)

dept_total <- hospital %>%
  group_by(Department) %>%
  summarise(Total = sum(Admissions))

ggplot(dept_total, aes(x = Department, y = Total, fill = Department)) +
  geom_bar(stat = "identity") +
  labs(title = "Total Patient Admissions by Department",
       x = "Department",
       y = "Total Admissions") +
  theme_minimal()

ggplot(hospital, aes(x = Month, y = Admissions, fill = Department)) +
  geom_bar(stat = "identity") +
  labs(title = "Monthly Admissions by Department",
       x = "Month",
       y = "Admissions") +
  theme_minimal()

treemap(
  dept_total,
  index = "Department",
  vSize = "Total",
  title = "Department-wise Patient Admissions"
)
ggplot(hospital, aes(x = Department, y = Admissions, fill = Department)) +
  geom_boxplot() +
  labs(title = "Admission Variability by Department",
       x = "Department",
       y = "Admissions") +
  theme_minimal()

ggplot(hospital, aes(x = Admissions)) +
  geom_histogram(binwidth = 20, fill = "skyblue", color = "black") +
  labs(title = "Distribution of Patient Admissions",
       x = "Admissions",
       y = "Frequency") +
  theme_minimal()
```









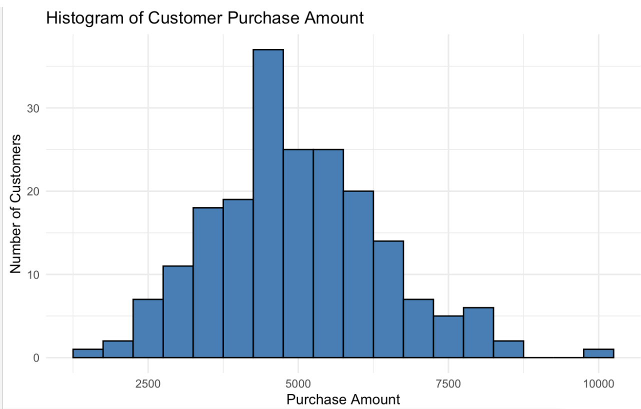


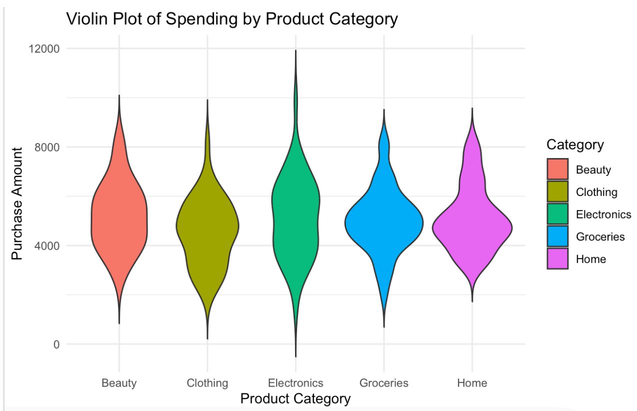
2. E-Commerce Customer Spending Behaviour

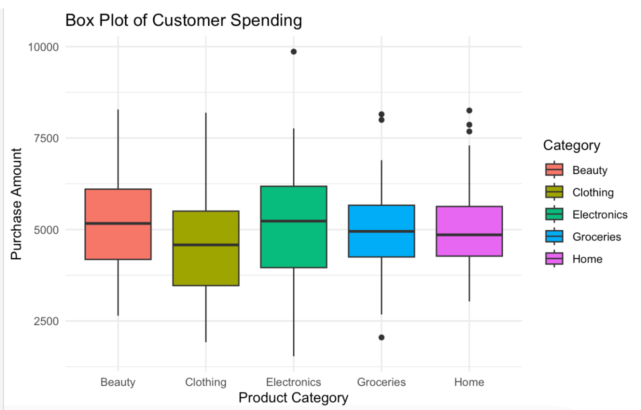
R program

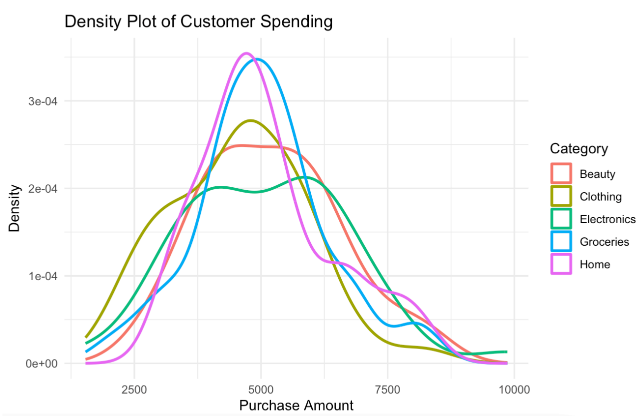
```
library(ggplot2)
library(dplyr)

set.seed(123)

customers <- data.frame(
  CustomerID = paste0("C", 1:200),
  PurchaseAmount = round(rnorm(200, mean = 5000, sd = 1500)),
  OrderFrequency = sample(1:20, 200, replace = TRUE),
  Category = sample(c("Electronics", "Clothing", "Groceries", "Home", "Beauty"),
                    200, replace = TRUE)
)

customers$PurchaseAmount[customers$PurchaseAmount < 500] <- 500

ggplot(customers, aes(x = PurchaseAmount)) +
  geom_histogram(binwidth = 500, fill = "steelblue", color = "black") +
  labs(title = "Histogram of Customer Purchase Amount",
```

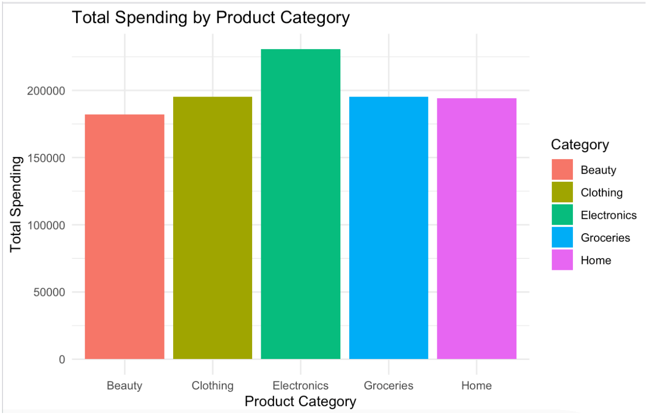
```r
      x = "Purchase Amount",
      y = "Number of Customers") +
  theme_minimal()

ggplot(customers, aes(x = Category, y = PurchaseAmount, fill = Category)) +
  geom_violin(trim = FALSE) +
  labs(title = "Violin Plot of Spending by Product Category",
      x = "Product Category",
      y = "Purchase Amount") +
  theme_minimal()

ggplot(customers, aes(x = Category, y = PurchaseAmount, fill = Category)) +
  geom_boxplot() +
  labs(title = "Box Plot of Customer Spending",
      x = "Product Category",
      y = "Purchase Amount") +
  theme_minimal()

ggplot(customers, aes(x = PurchaseAmount, color = Category)) +
  geom_density(size = 1) +
  labs(title = "Density Plot of Customer Spending",
      x = "Purchase Amount",
      y = "Density") +
  theme_minimal()

category_total <- customers %>%
  group_by(Category) %>%
  summarise(TotalSpending = sum(PurchaseAmount))

ggplot(category_total, aes(x = Category, y = TotalSpending, fill = Category)) +
  geom_bar(stat = "identity") +
  labs(title = "Total Spending by Product Category",
      x = "Product Category",
      y = "Total Spending") +
  theme_minimal()
```









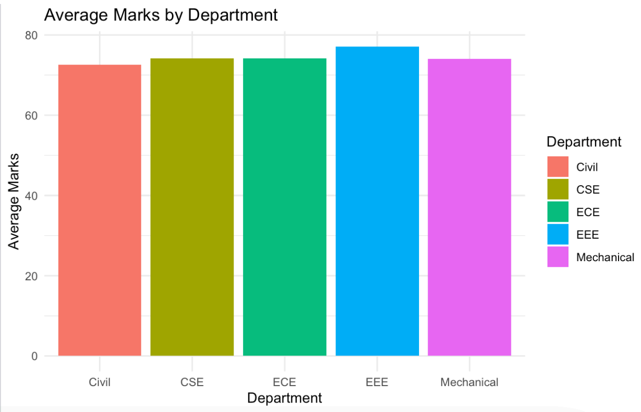


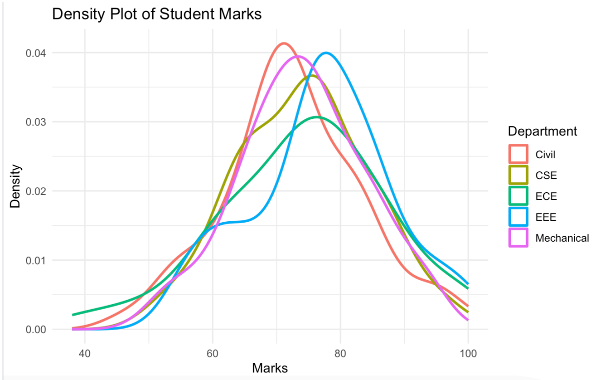
3. University Examination Performance Dashboard

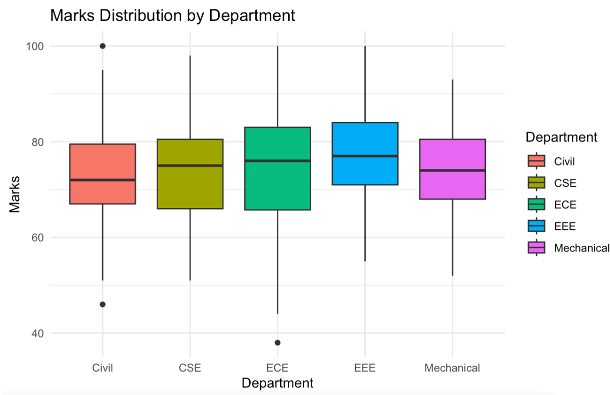
R program

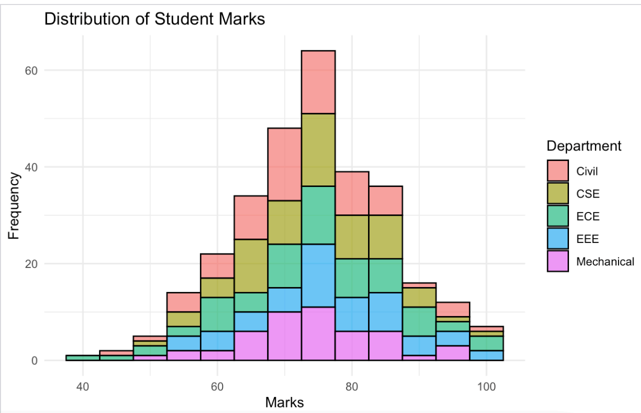
```
library(ggplot2)
library(dplyr)

set.seed(123)

students <- data.frame(
  Department = sample(c("CSE", "ECE", "EEE", "Mechanical", "Civil"), 300, replace = TRUE),
  Class = sample(c("A", "B", "C"), 300, replace = TRUE),
  Marks = round(rnorm(300, mean = 75, sd = 12))
)

students$Marks[students$Marks > 100] <- 100
students$Marks[students$Marks < 0] <- 0

dept_avg <- students %>%
  group_by(Department) %>%
  summarise(AverageMarks = mean(Marks))

class_avg <- students %>%
  group_by(Class) %>%
  summarise(AverageMarks = mean(Marks))

ggplot(dept_avg, aes(x = Department, y = AverageMarks, fill = Department)) +
  geom_bar(stat = "identity") +
  labs(title = "Average Marks by Department",
       x = "Department",
       y = "Average Marks") +
  theme_minimal()

ggplot(students, aes(x = Department, y = Marks, fill = Department)) +
  geom_boxplot() +
  labs(title = "Marks Distribution by Department",
       x = "Department",
       y = "Marks") +
  theme_minimal()

ggplot(students, aes(x = Marks, fill = Department)) +
  geom_histogram(binwidth = 5, color = "black", alpha = 0.7) +
  labs(title = "Distribution of Student Marks",
       x = "Marks",
       y = "Frequency") +
  theme_minimal()

ggplot(students, aes(x = Marks, color = Department)) +
  geom_density(size = 1) +
  labs(title = "Density Plot of Student Marks",
       x = "Marks",
       y = "Density") +
  theme_minimal()
```

```
ggplot(class_avg, aes(x = Class, y = AverageMarks, fill = Class)) +
  geom_bar(stat = "identity") +
  labs(title = "Average Marks by Class",
       x = "Class",
       y = "Average Marks") +
  theme_minimal()
```







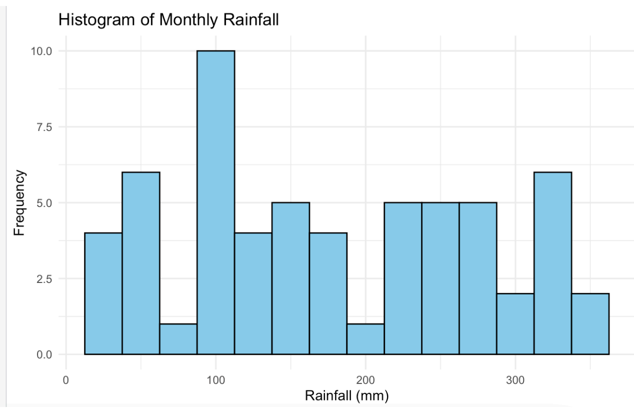


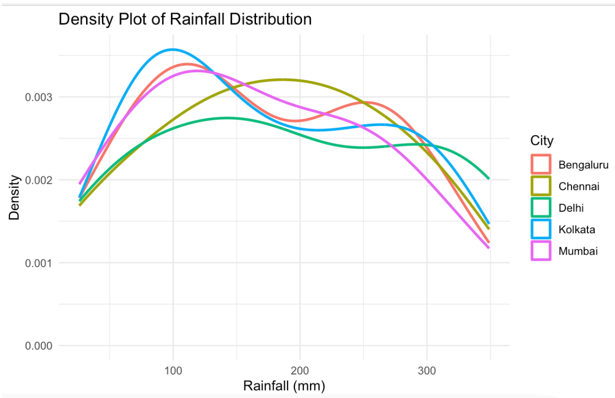
4. Climate and Rainfall Distribution Study

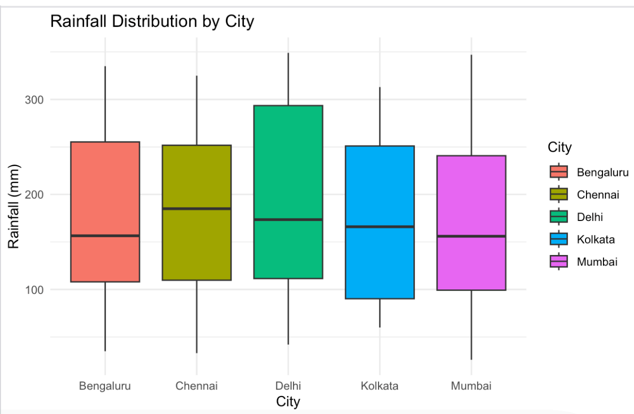
R program

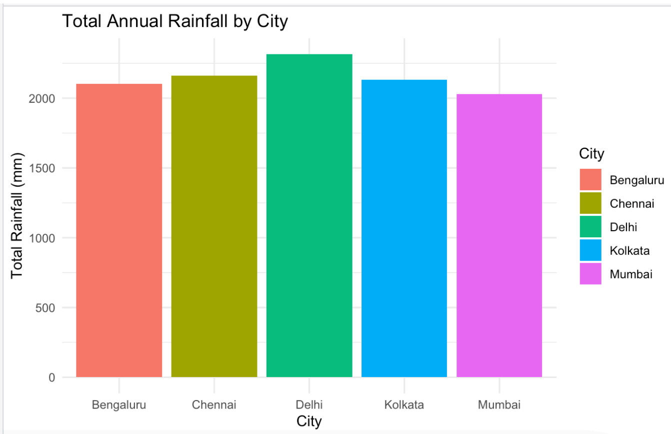
```
library(ggplot2)
library(dplyr)

set.seed(123)
```

```
climate <- data.frame(
  City = rep(c("Chennai", "Mumbai", "Delhi", "Kolkata", "Bengaluru"), each = 12),
  Month = rep(month.abb, 5),
  Rainfall = sample(20:350, 60, replace = TRUE),
  Temperature = round(runif(60, 18, 40), 1),
  Humidity = sample(50:95, 60, replace = TRUE)
)

city_rain <- climate %>%
  group_by(City) %>%
  summarise(TotalRainfall = sum(Rainfall))

ggplot(climate, aes(x = Rainfall)) +
  geom_histogram(binwidth = 25, fill = "skyblue", color = "black") +
  labs(title = "Histogram of Monthly Rainfall",
       x = "Rainfall (mm)",
       y = "Frequency") +
  theme_minimal()

ggplot(climate, aes(x = Rainfall, color = City)) +
  geom_density(size = 1) +
  labs(title = "Density Plot of Rainfall Distribution",
       x = "Rainfall (mm)",
       y = "Density") +
  theme_minimal()

ggplot(climate, aes(x = City, y = Rainfall, fill = City)) +
  geom_boxplot() +
  labs(title = "Rainfall Distribution by City",
       x = "City",
       y = "Rainfall (mm)") +
  theme_minimal()

ggplot(city_rain, aes(x = City, y = TotalRainfall, fill = City)) +
  geom_bar(stat = "identity") +
  labs(title = "Total Annual Rainfall by City",
       x = "City",
       y = "Total Rainfall (mm)") +
  theme_minimal()
```





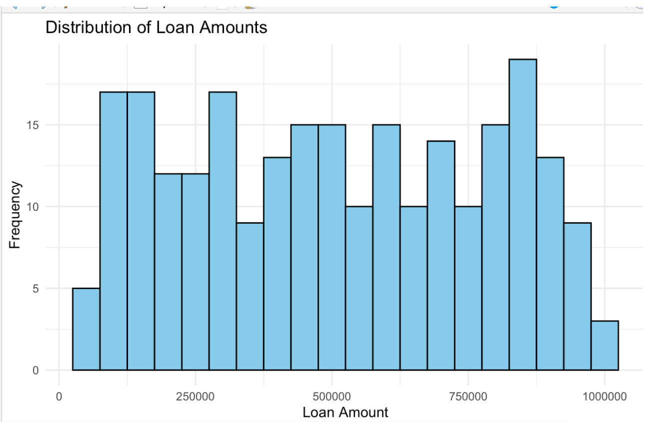


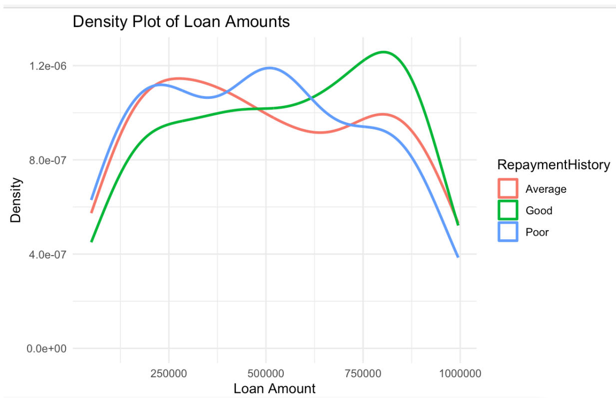


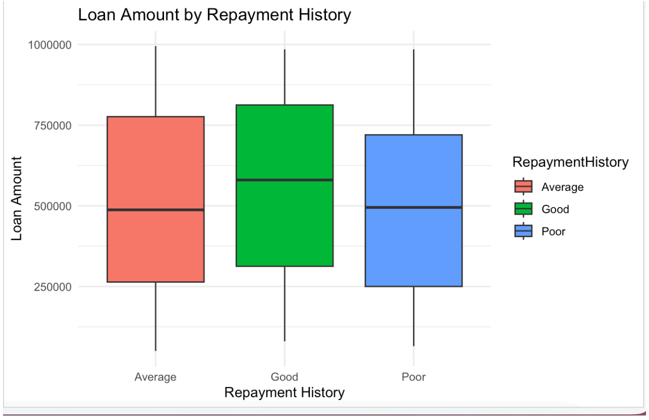
5. Bank Loan Risk Assessment

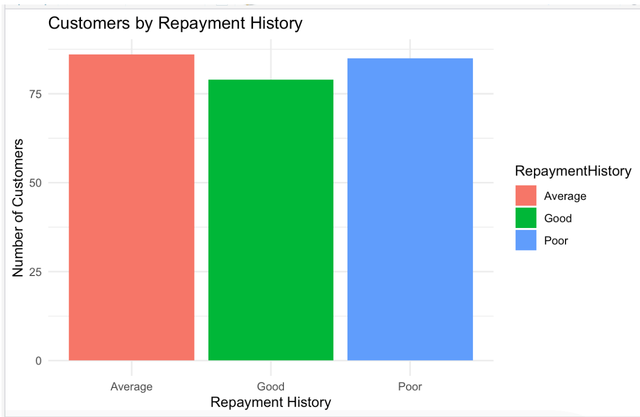
R program

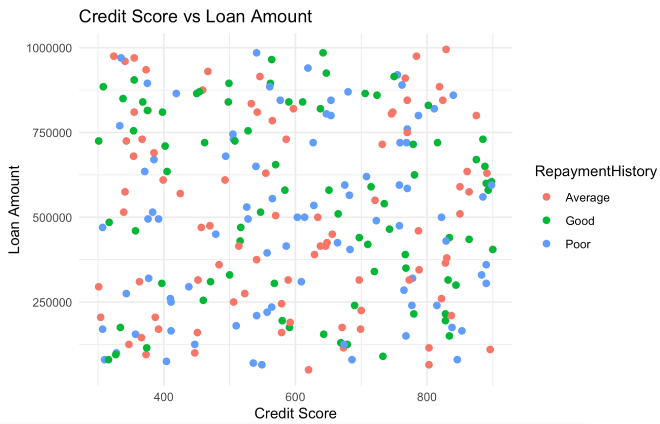
```
library(ggplot2)
library(dplyr)

set.seed(123)

loans <- data.frame(
  CustomerID = paste0("C", 1:250),
  LoanAmount = sample(seq(50000, 1000000, by = 5000), 250, replace = TRUE),
  Income = sample(seq(20000, 150000, by = 1000), 250, replace = TRUE),
  CreditScore = sample(300:900, 250, replace = TRUE),
  RepaymentHistory = sample(c("Good", "Average", "Poor"), 250, replace = TRUE)
)

repayment_count <- loans %>%
  group_by(RepaymentHistory) %>%
  summarise(Customers = n())

ggplot(loans, aes(x = LoanAmount)) +
  geom_histogram(binwidth = 50000, fill = "skyblue", color = "black") +
  labs(title = "Distribution of Loan Amounts",
       x = "Loan Amount",
       y = "Frequency") +
  theme_minimal()

ggplot(loans, aes(x = LoanAmount, color = RepaymentHistory)) +
  geom_density(size = 1) +
  labs(title = "Density Plot of Loan Amounts",
       x = "Loan Amount",
       y = "Density") +
  theme_minimal()
ggplot(loans, aes(x = RepaymentHistory, y = LoanAmount, fill = RepaymentHistory)) +
  geom_boxplot() +
  labs(title = "Loan Amount by Repayment History",
       x = "Repayment History",
       y = "Loan Amount") +
  theme_minimal()
ggplot(repayment_count, aes(x = RepaymentHistory, y = Customers, fill = RepaymentHistory)) +
  geom_bar(stat = "identity") +
  labs(title = "Customers by Repayment History",
       x = "Repayment History",
       y = "Number of Customers") +
  theme_minimal()

ggplot(loans, aes(x = CreditScore, y = LoanAmount, color = RepaymentHistory)) +
  geom_point(size = 2) +
  labs(title = "Credit Score vs Loan Amount",
       x = "Credit Score",
       y = "Loan Amount") +
  theme_minimal()
```






Loan Amount by Repayment History
1000000
750000
500000
250000
Loan Amount
Average
Good
Poor
Repayment History
RepaymentHistory
Average
Good
Poor


Customers by Repayment History
75
50
25
0
Number of Customers
Average
Good
Poor
Repayment History
RepaymentHistory
Average
Good
Poor


Credit Score vs Loan Amount
1000000
750000
500000
250000
Loan Amount
400
600
800
Credit Score
RepaymentHistory
Average
Good
Poor